ROHIT JINDAL

Senior Data Scientist - Applied AI

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PROFESSIONAL SUMMARY

Senior Data Scientist specializing in **Applied AI, Generative AI, and large-scale decision systems**, with 5+ years of experience delivering measurable business outcomes. Proven track record of deploying **production-grade ML and AI solutions generating multi-million-dollar impact**, including automation of complex enterprise workflows, optimization of large-scale operational systems, and development of intelligent AI agents using modern **RAG and transformer-based architectures**.

Combines strong foundations in **statistics, machine learning, mathematics, and optimization** with systems thinking to translate ambiguous business problems into **scalable, high-impact AI products** used in production environments.

EXPERIENCE

TARGET — Bangalore, India

Senior Data Scientist - NLP & Gen AI Team — 08/2023 - Present

- Built and deployed a Siamese-BERT classification system to automate PCC code assignment, reducing processing time from days to minutes, achieving approximately 78% accuracy, and generating around $1M in annual operational savings.
- Generated $2.5M impact by developing a Markov-chain inventory optimization framework. Integrated with ML pipelines and XGBoost models. Achieved 78% recall, enhancing inventory planning decisions.
- Developed an AI agent using RAG for finance, incorporating FAISS/Qdrant for automated document validation and compliance, streamlining recommendation workflows, and reducing manual review.
- Implemented LLM observability pipelines with LangSmith for GenAI system monitoring, debugging, and optimization.

Indian Railways — Delhi, India

Assistant Manager (Executive Officer) — 2019 - 2021

- Developed ML models (Decision Trees, SVM, Random Forest) to predict 28-day concrete compressive strength, reducing engineering analysis time from days to minutes for a ₽1,600-crore railway infrastructure project, accelerating approvals and engineering decisions.
- Led execution of INR 3 crore infrastructure projects, managing a team of three site engineers and coordinating with government stakeholders to secure multi-crore project approvals.

State Street Global Advisors (Apprenticeship) — Bangalore, India

Quantitative Analyst (Financial ML)

- Developed a novel equity factor "Short Interest Surprise," improving portfolio performance by approximately 7% across short-term horizons.
- Built large-scale backtesting pipelines on millions of equity records using Spark and R on Databricks, generating approximately 2.3% alpha and over $1 million potential returns for high-net-worth individual client portfolios.
- Designed client-facing analytics dashboards to monitor portfolio performance and factor model insights.

Mastercard AI Garage (Internship) — Gurugram, India

Data Scientist 2

- Improved fraud detection pipeline recall by 7% through drift-detection modeling on imbalanced datasets.
- Improved fraud detection recall by 7% through drift-aware modeling techniques on imbalanced transaction datasets.
- Applied machine learning models on 15M+ financial transactions using Spark, enhancing fraud risk insights and detection accuracy.
- Research contribution accepted at ICAIF 2021 (ACM Conference on AI in Finance).
- Applied ML models on 15M+ transactions using Spark to enhance fraud risk insights; Submitted Research paper at ICAIF 2021.

Education

Indian Institute of Science (IISc) — Bangalore, India

Masters in Data & Analytics | Score: **8.2** / 10 — 2021 - 2023

- Econometrics, Statistics, Mathematics, Algorithms

Delhi College of Engineering — Delhi, India

Bachelors of Technology | Percentage: **80** / 100 — 2015 - 2019

Education

University of Cambridge & Bits Pilani
Cambridge, United Kingdom

Sustainability & IP Data

2023 - 2024

- IP & Sustainability in Data Science at King's College, Cambridge

DPS, Paschim Vihar
New Delhi, India

Class XII & Class X | Percentage: **95** / 100

- Class XII School Topper - 95%
- Class X School Topper - 9.6 CGPA

TECH STACK

Language & Systems: Python, SQL, R, PySpark, Cloud AWS

Gen AI: LangChain, LangGraph, LangSmith, Vector DB, Advanced RAG

ML & AI: XGBoost, Transformers, Deep Learning

Cloud & ML Ops: AWS, GCP, Docker

AI SYSTEMS & KEY PROJECTS

Hybrid Email Intelligence System

Python, spaCy, Scikit-learn, Sentence-Transformers, OpenAI, Streamlit.

- Developed a Hybrid Email Intelligence System for Pegasystems. Created NLP pipeline using TF-IDF, Logistic Regression, Sentence Embeddings, KNN to reduce LLM use. Achieved 92% accuracy; cut LLM costs by 72%. Built with Streamlit, FastAPI, with confidence calibration and human validation.
- Built a weighted ensemble classifier (LLM few-shot + TF-IDF/Log Reg + Sentence-BERT/KNN) with cosine-similarity voting, achieving 92% intent accuracy across 6 categories while reducing LLM API costs by approximately 72% through selective fallback.
- Engineered a multi-strategy field extractor combining spaCy NER, regex-based date/order-ID parsing, and keyword-action mapping with LLM fallback only for missing critical fields — outputting CRM-ready structured JSON.
- Designed confidence-based auto-routing logic using intent thresholds, classifier agreement, urgency, and sentiment scoring to flag low-confidence emails for human review.
- Deployed on cloud using Streamlit as frontend, batch processing, intent distribution charts, and downloadable exports for stakeholder walkthroughs.

Competitor Intelligence-Driven Restaurant Success Prediction System

- Developed a Random Forest model with 79% OOB accuracy and 0.82 F1-score. Identified top market drivers impacting restaurant success. Provided insights for new restaurant placement, underperforming locations, and revenue optimization to Store Strategy and Real Estate teams.
- Collaborated with Competitive Intelligence to unify geospatial data for restaurant modeling. Integrated datasets: competitor footfall, demographics, POI density, sales. Engineered location features using K-Means clustering to assess market saturation, cannibalization, and competitor proximity.
- Delivered actionable insights to Store Strategy and Real Estate teams to inform new restaurant placement, underperforming location intervention, and revenue optimization decisions

LEADERSHIP & MENTORING

AI Educator & Course Designer

Leadership Impact

- Developed and taught 700+ hours of AI curriculam for 100+ professionals, covering ML, Deep Learning, NLP.
- Guest speaker at REVA and SST, lecturing on applied ML, neural networks, and AI careers.

LEADERSHIP & MENTORING

Academic Achievements

Key Achievements

- GATE All India Rank - 079
- JEE Mains 99.99 percentile
- Class Representative & Batch Placement Coordinator, IISc Bangalore (Full Batch Placed)
- Class Representative, B.Tech (DTU)
- Class X and Class XII - Academic Topper - 98%